TOPIC 29/60

Prompting vs. Fine-Tuning

> The Battle of Adaptation.
>
> How do you teach a general-purpose AI to perform your specific, highly-specialized enterprise task? There are two radically different approaches.

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THE CORE DILEMMA

The Blank Slate Model

Base LLMs (like Llama 3 or GPT-4 base) are trained on the entire internet. They know a little about everything, but they don't know exactly what *you* want them to do right now.

The Task: Extract JSON from medical text.

Without Adaptation

The model might summarize the text, write a poem about doctors, or output invalid JSON.

You have two options to fix this:

1. Prompting

Tell it exactly what to do at runtime.

2. Fine-Tuning

Rewire its brain permanently.

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METHOD 1

Prompting (In-Context Learning)

Prompting modifies the model's behavior **temporarily** by providing instructions and examples in the context window. The underlying neural network weights *do not change*.

THE API REQUEST

```
System: "You are a JSON extractor. Output ONLY valid JSON."

User: "Extract from this: 'Patient John is 45.'"

Assistant: "{ \"name\": \"John\", \"age\": 45 }"

User: "Extract from this: 'Jane broke her arm.'"

Assistant: { "name": "Jane", "injury": "broken arm" }
```

Status: Fast, agile, but uses lots of tokens every single time.

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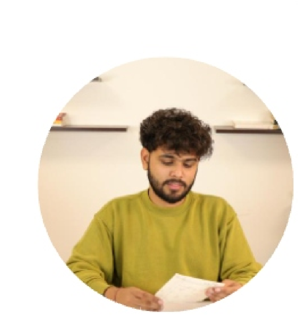

METHOD 2

Fine-Tuning

Fine-tuning involves taking a pre-trained model and running **gradient descent** on a curated dataset. The model's actual mathematical weights are permanently updated.

1. Prepare Dataset

Create 10,000 pairs of Medical Text ➔ JSON.

➔

2. Train Model

Update neural weights over several hours on GPUs.

THE NEW API REQUEST

```
User: "Jane broke her arm."
Assistant: { "name": "Jane", "injury": "broken arm" }
```

Status: No instructions needed. The format is baked into its "muscle memory."

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UNDERSTANDING THE DIFFERENCE

The Employee Analogy

Prompting

Hiring a brilliant temp worker.

Every morning, you hand them a 10-page instruction manual on how to do their job. They read it, do the job perfectly, and go home.

Pros: No training time.

Cons: You have to pay them to read the manual every single day.

Fine-Tuning

Sending them to a bootcamp.

You pay them to attend a 4-week intensive training program. When they return, they instantly know how to do the job without any manuals.

Pros: Fast daily execution. Cheap.

Cons: High upfront cost and time.

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THE BARRIER TO ENTRY

The Data Threshold

The biggest deciding factor between the two approaches is how much high-quality data you possess.

Prompting Needs:
0 to 10 examples (Few-Shot)

Fine-Tuning Needs:
500 to 50,000+ examples

Warning: Fine-tuning on bad data (or too little data) will cause "Catastrophic Forgetting" where the model gets dumber and loses its base knowledge.

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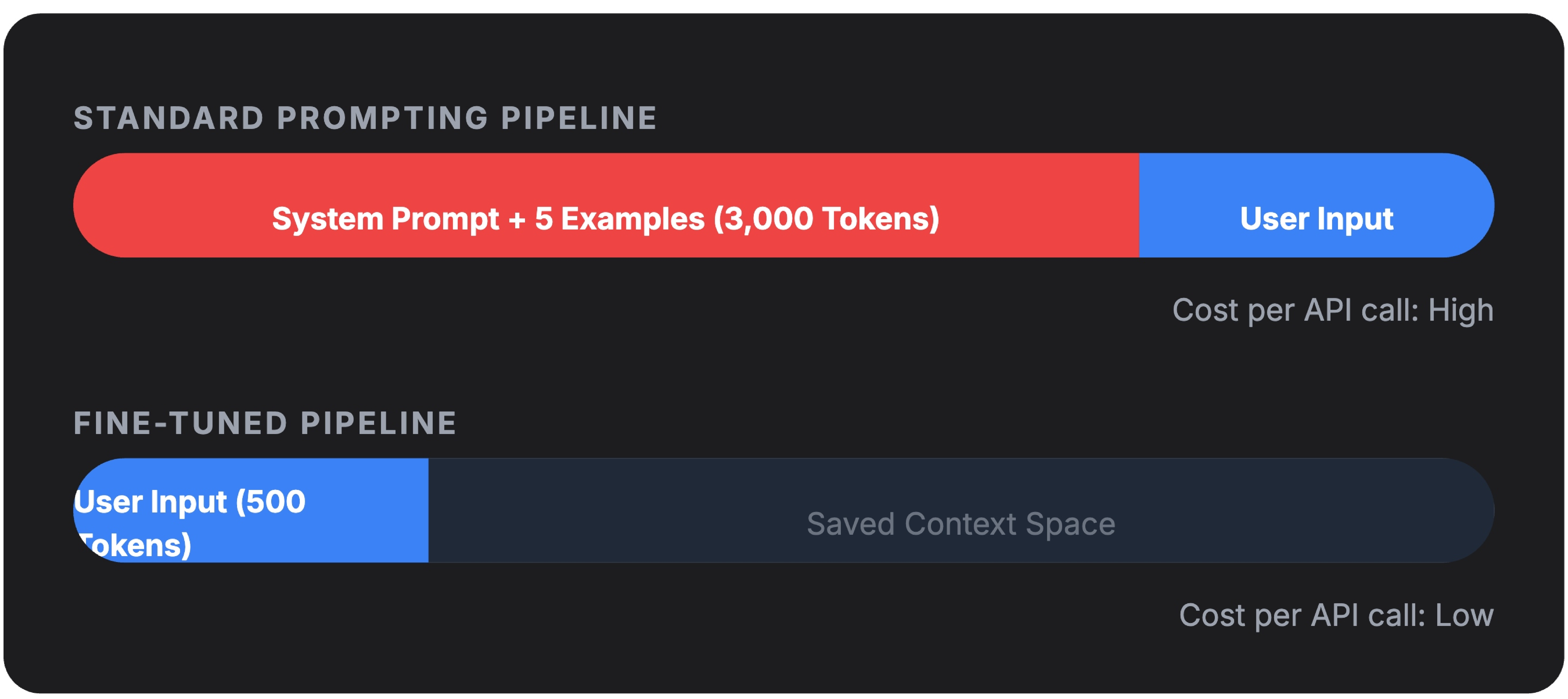
TOKEN ECONOMICS

Saving the Context Window

Every time you send a prompt, you pay for the input tokens. If your task requires a 2,000-word instruction set to perform correctly, prompting becomes incredibly expensive at scale.



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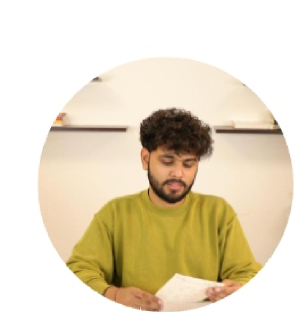

THE GRAND MISCONCEPTION

Don't Fine-Tune for Knowledge

The biggest mistake engineers make is trying to fine-tune a model to teach it *new facts* (e.g., fine-tuning on company HR documents).

❌ Fine-Tuning

Neural networks are lossy compressors. If you fine-tune a model on an HR doc, it will *memorize* it poorly. It will hallucinate details and you cannot easily delete the document later.

Bad for Fact Retrieval

✅ RAG (Prompting)

Retrieval-Augmented Generation searches a database, grabs the exact document, and pastes it into the prompt. The LLM reads it perfectly and grounds its answer.

Perfect for Fact Retrieval

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SUMMARY

The Decision Matrix

GOAL	USE PROMPTING / RAG	USE FINE-TUNING
Adding New Facts	✅ **Yes (RAG)**	❌ No
Enforcing strict JSON format	Okay, but uses tokens	✅ **Best approach**
Mimicking Brand Voice	Works with few-shot	✅ **Perfect (Style transfer)**
Low upfront budget	✅ **Instant & Free to start**	❌ Expensive compute

Golden Rule: Always try Prompting first. Only Fine-Tune if Prompting fails.

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The Hybrid Future

BEST OF BOTH WORLDS

🚀 Most enterprise systems use **both.**

⚙️ Fine-tune a smaller model (SLM) to understand formatting, tone, and reasoning style.

🔍 Use RAG (Prompting) to inject real-time database facts into that fine-tuned model.

💡 Result: High accuracy, low hallucination, cheap token costs.

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